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A Polynomial Zero Attracting Affine Projection Algorithm for Sparse System Identification

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Abstract: A polynomial zero attracting affine projection algorithm is prosed in this work for sparse system identification. The existing zero attracting algorithms based on affine projection algorithm for sparse system identification, consider an objective function which is a combination of an l_2-norm and an approximation of l_0-norm. The difference between these algorithms is the approximation of the l_0-norm. In order to further improve the zero attraction capability of sparse adaptive algorithms based on affine projection algorithm, a polynomial is used to approximate the l_0-norm. Simulation results verify that the proposed algorithm is effective for sparse system identification.

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Keywords: sparse system identification, affine projection algorithm, polynomial, zero attracting.

1. INTRODUCTION

Estimating and balancing mobile communication channels is a hot topic, but usually it is a sparse system. A sparse system is a system with a majority of coefficients of zero or near zero for pulse response. Fortunately, it has been proposed in (Donoho, D L. Lustig, M. Santos, J M. Pauly, 2006) that sparsity can be best represented by l_0 norm, in which constraint the sparsest solution is acquirable. However, since minimizing the l_0 norm is a Non-Polynomial (NP) hard problem, attempts have been made to employ other functions to instead of the l_0 norm. Some algorithms for sparse system identification based on Least Mean Square(LMS) have been proposed in(Shi and Zhao, 2018) (Chen, Gu and Hero, 2009).Suppose that most of the smaller coefficients actually correspond to zero terms, so in the zero attraction mechanism, the smaller coefficients of the model are forced to zero. When identifying sparse systems, the zero attractor can speed up the convergence, due to the ability of promoting sparsity in the filter coefficients during the update process. The Zero-Attracting LMS (ZA-LMS) and the reweighted Zero-Attracting LMS (RZA-LMS), which proposed in (Chen, Gu and Hero, 2009), generate a zero attractor by combining an l_1 norm penalty with the quadratic LMS cost function. In order to further improve the filtering performance, the reweighted zero attractor has been used in the RZA-LMS algorithm. The l_0-LMS algorithm, proposed in (Gu, Jin and Mei, 2009), has been shown to provide improved convergence over the ZA-LMS and RZA-LMS algorithms. The key to this algorithm is to use the exponential function to approximate the l_0-norm penalty. Recently the polynomial Zero-Attracting LMS (PZA-LMS) algorithm, which approximates the l_0-norm penalty using an α-order polynomial, has been proposed in (Maheshwari and George, 2016). In the sparse conditions without compromising zero-attraction intensity, the PZA-LMS algorithm improves the filtering performance.

Compared to algorithms based on Least Mean Square, the Affine Projection Algorithm (APA) (Diniz, 2013) reuses old data resulting in a faster convergence. Recently, the Zero-Attracting Affine Projection Algorithm (ZA-APA) and the Reweighted Zero-Attracting Affine Projection Algorithm (RZA-APA) has been introduced in (Ran Meng, de Lamare and Nascimento, 2011). In order to generate a zero attractor in the update function, the ZA-APA combines a l_1 norm penalty on the coefficients with a standard APA cost function. It is crucially, the ZA-APA does not distinguish between zero taps and non-zero taps. So, the performance of ZA-APA deteriorates when applied to less sparse systems. To solve this problem, the author in (Ran Meng, de Lamare and Nascimento, 2011) proposed the RZA-APA, which a reweighted zero attractor is used. An l_0-APA algorithm has been proposed in (Pelekanakis and Chitre, 2013) for sparse system identification, which approximates the l_0-norm using an exponential function.

Motivated by the algorithms pointed in above, the main idea to deal with the sparse systems identify is use the zero attraction mechanism. So, adding a penalty to the cost function is necessary, which can represent the sparsity of sparse systems. Although, l_0 norm is the best chose, it's a NP

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hard problem to minimizing the l_0 norm. To circumvent the NP hard problem, choosing other functions to approximate the l_0 norm is an effective way. So as to further improve the performance for APA algorithm, during identifying the sparse systems, this paper proposes a polynomial ZA-APA (PZA-APA) algorithm, which tries to approximate the l_0 norm using an α-order polynomial (Maheshwari and George, 2016).

The rest of the brief is organized as follows: The proposed algorithm is introduced in Section 2. And a detailed simulation study is carried out in Section 3 to test the effectiveness of the proposed algorithm and the concluding remarks are drawn in Section 4.

2. PROPOSED ALGORITHM

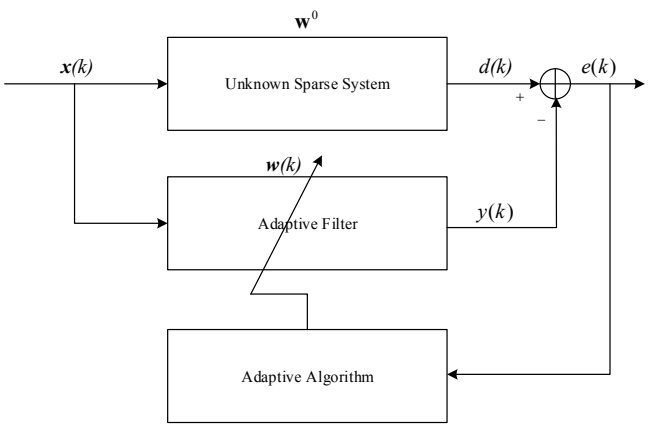

Fig. 1. Block diagram of sparse system identification based on adaptive algorithm.

During identifying a sparse system, the input signal $\boldsymbol{x}(k) \in \mathbb{R}^{N+1}$ at time $k \in \mathbb{Z}$ does two things, which is shown in Fig.1. One is to generate the desired signal $d(k) \in \mathbb{R}$, which is the output of the unknown sparse system. The other is to produce an output estimate $y(k) \triangleq \boldsymbol{x}^T(k)\boldsymbol{w}(k)$, where $\boldsymbol{w}(k) \in \mathbb{R}^{N+1}$ is the coefficient vector. The difference between the desired signal $d(k)$ and the output estimate $y(k)$ is the error signal $e(k) \triangleq d(k) - y(k)$. Here $\mathbf{w}^0 \in \mathbb{R}^{N+1}$ is the coefficient vector of the unknown sparse system.

The standard APA, reuses previous input vectors to increase convergence speed. We assume the last $L+1 \in \mathbb{N}$ input signal vectors as follows:

$$\boldsymbol{X}(k) \triangleq [\boldsymbol{x}(k), \boldsymbol{x}(k-1) \cdots \boldsymbol{x}(k-L)] \in \mathbb{R}^{(N+1)\times(L+1)},$$

where $\boldsymbol{X}(k)$ is a matrix and $L+1 \in \mathbb{N}$ denotes the APA order. Similarly, we are able to define the desired vector $\boldsymbol{d}(k)$ and the error vector $\boldsymbol{e}(k)$.

$$\boldsymbol{d}(k) \triangleq [d(k), d(k-1) \cdots d(k-L)]^T \in \mathbb{R}^{L+1},$$

$$\boldsymbol{e}(k) \triangleq \boldsymbol{d}(k) - \boldsymbol{X}^T(k)\boldsymbol{w}(k) \tag{1}$$

Geometrically, the updating process of the APA yields a coefficient vector $\boldsymbol{w}(k+1)$ which is as close as possible to $\boldsymbol{w}(k)$ (minimum perturbation criterion in the Euclidean distance sense) and whose associated a posteriori error is equal to zero, i.e., $\boldsymbol{d}(k) - \boldsymbol{X}^T(k)\boldsymbol{w}(k+1) = 0$. Mathematically, the optimization problem solved by the APA is

$$\text{minimize } \|\boldsymbol{w}(k+1) - \boldsymbol{w}(k)\|_2^2$$

$$\text{subject to } \boldsymbol{d}(k) - \boldsymbol{X}^T(k)\boldsymbol{w}(k+1) = 0.$$

After using the Lagrange multiplier, the problem is transformed into minimize the cost function $J_1(k)$.

$$J_1(k) = \|\boldsymbol{w}(k+1) - \boldsymbol{w}(k)\|_2^2 + \boldsymbol{\lambda}^T(k)[\boldsymbol{d}(k) - \boldsymbol{X}^T(k)\boldsymbol{w}(k+1)] \tag{2}$$

where $\boldsymbol{\lambda}(k) \in \mathbb{R}^{L+1}$ is the vector of Lagrange multipliers. Then, differentiate $J_1(k)$ with $\boldsymbol{w}(k+1)$ and $\boldsymbol{\lambda}(k)$ respectively. After that, we assume the resulting expressions to zero $\nabla J_1(k) = 0$, thus yielding the updating equation of the APA:

$$\boldsymbol{w}(k+1) = \boldsymbol{w}(k) + \boldsymbol{X}(k)(\boldsymbol{X}^T(k)\boldsymbol{X}(k))^{-1}\boldsymbol{e}(k) \tag{3}$$

There is a matrix inversion in the updating equation, so there may be numerical instability. To avoid this, we use the regularization factor $\delta \in \mathbb{R}_+$, which $\delta \ll 1$. In addition, a convergence factor (step-size) $\mu \in \mathbb{R}_+$ with $\mu < 1$, is also employed to relax the constraint of the optimization problem, thus enabling smaller steps in the updating process. Therefore, the APA is commonly implemented as (Diniz, 2013):

$$\boldsymbol{w}(k+1) = \boldsymbol{w}(k) + \mu\boldsymbol{X}(k)(\boldsymbol{X}^T(k)\boldsymbol{X}(k) + \delta I)^{-1}\boldsymbol{e}(k) \tag{4}$$

Compared to APA, the ZA-APA, RZA-APA and l_0-APA algorithms consider an objective function which is a combination of an l_2-norm and an approximation of l_0-norm. It may be noted that the major difference between the objective functions of the ZA-APA, RZA-APA and l_0-APA algorithms reported recently are in the type of approximation of l_0-norm employed in the design of the objective function. In ZA-APA, the l_0-norm is approximated as an l_1-norm and is given by $\xi_{l_0}(k) = \|\boldsymbol{w}(k)\|_1$ where as in RZA-LMS it is given by $\xi_{l_0}(k) = \sum_{i=1}^{N+1} \log(1 + |w_i(k)| / \varepsilon)$ where $\varepsilon > 0$ is the shrinkage magnitude and $w_i(k)$ is the ith dimension coefficient of $\boldsymbol{w}(k)$. The l_0-norm approximation employed in l_0-APA is given as $\xi_{l_0}(k) = \sum_{i=1}^{N+1}(1 - \exp(-\beta|w_i(k)|))$ where β is the zero attraction intensity parameter. A function that can approximate to l_0 norm has been proposed in (Maheshwari and George, 2016).

$$\xi_{l_0}(k) = \sum_{i=1}^{N+1} \begin{cases} \dfrac{|w_i(k)|}{(1+|w_i(k)|)^{\alpha}}, & |w_i(k)| \le \dfrac{1}{\alpha - 1} \\ \dfrac{|\alpha - 1|^{\alpha-1}}{|\alpha|^{\alpha}}, & elsewhere \end{cases} \tag{5}$$

Summarizing the observations in (Maheshwari and George, 2016), it can be pointed out that the proposed PZA-APA algorithm can offer zero attraction to very near to zero coefficients of the weight vector without significant reduction in the magnitude of other near zero coefficients. The

optimization problem of the proposed PZA-APA algorithm is given by

$$\text{minimize } \|\boldsymbol{w}(k+1)-\boldsymbol{w}(k)\|_2^2+\gamma\xi_{l_0}(k)$$

$$\text{subject to } \boldsymbol{d}(k)-\boldsymbol{X}^T(k)\boldsymbol{w}(k+1)=0\,.$$

where $\gamma\in\mathbb{R}_+$ is a nonnegative parameter that determines the weight given to the l_0 norm penalty. In order to solve this optimization problem, we use the Lagrange multiplier. Then the problem is transformed into minimize the cost function $J_2(k)$.

$$\begin{aligned}J_2(k)&=\|\boldsymbol{w}(k+1)-\boldsymbol{w}(k)\|_2^2\\&+\boldsymbol{\lambda}^T(k)[\boldsymbol{d}(k)-\boldsymbol{X}^T(k)\boldsymbol{w}(k+1)]\\&+\gamma\xi_{l_0}(k)\end{aligned}\tag{6}$$

Differentiate $J_2(k)$ with respect to $\boldsymbol{w}(k+1)$ and $\boldsymbol{\lambda}(k)$, and equal the resulting expressions to zero $\nabla J_2(k)=0$, thus yielding

$$\boldsymbol{w}(k+1)=\boldsymbol{w}(k)+\boldsymbol{X}(k)\frac{\boldsymbol{\lambda}(k)}{2}-\frac{\gamma}{2}\nabla\boldsymbol{\xi}_{l_0}(k)\tag{7}$$

$$\boldsymbol{d}(k)=\boldsymbol{X}^T(k)\boldsymbol{w}(k+1)\tag{8}$$

where $\nabla\boldsymbol{\xi}_{l_0}(k)\in\mathbb{R}^{N+1}$

$$\frac{\partial\xi_{l_0}(k)}{\partial w_i(k)}=\begin{cases}\dfrac{\operatorname{sgn}[w_i(k)][1-(\alpha-1)|w_i(k)|]}{[1+|w_i(k)|]^{\alpha+1}}, & |w_i(k)|\le\dfrac{1}{\alpha-1}\\ 0, & elsewhere\end{cases}$$

Let $g_i(k)=\partial\xi_{l_0}(k)/\partial w_i(k)$, then $\nabla\boldsymbol{\xi}_{l_0}(k)$ can be written in the following form $\nabla\boldsymbol{\xi}_{l_0}(k)=[g_1(k),g_2(k),\cdots,g_{N+1}(k)]$.

Then, the left-multiplication of eq. (7) by $\boldsymbol{X}^T(k)$ and the substitution of eq. (8) into the resulting equation generates (assuming that $\boldsymbol{X}^T(k)\boldsymbol{X}(k)$ is invertible)

$$\begin{aligned}\frac{\boldsymbol{\lambda}(k)}{2}&=(\boldsymbol{X}^T(k)\boldsymbol{X}(k))^{-1}\boldsymbol{e}(k)\\&+\frac{\gamma}{2}(\boldsymbol{X}^T(k)\boldsymbol{X}(k))^{-1}\boldsymbol{X}^T(k)\nabla\boldsymbol{\xi}_{l_0}(k)\end{aligned}\tag{9}$$

Substituting eq. (3) into eq. (1) leads to the updating equation of PZA-APA

$$\begin{aligned}\boldsymbol{w}(k+1)&=\boldsymbol{w}(k)+\boldsymbol{X}(k)(\boldsymbol{X}^T(k)\boldsymbol{X}(k))^{-1}\boldsymbol{e}(k)\\&+\frac{\gamma}{2}[\boldsymbol{X}(k)(\boldsymbol{X}^T(k)\boldsymbol{X}(k))^{-1}\boldsymbol{X}^T(k)-I]\nabla\boldsymbol{\xi}_{l_0}(k)\end{aligned}\tag{10}$$

In practice, a regularization factor $\delta\in\mathbb{R}_+$ and $\delta\ll1$ is usually employed to avoid numerical instability due to matrix inversion. In addition, a convergence factor (step-size) $\mu\in\mathbb{R}_+$ and $\mu<1$ is also used to relax the constraint of the optimization problem, thus enabling smaller steps in the updating process. Therefore, the PZA-APA algorithm can be written in the following form:

$$\begin{aligned}\boldsymbol{w}(k+1)&=\boldsymbol{w}(k)\\&+\mu\boldsymbol{X}(k)(\boldsymbol{X}^T(k)\boldsymbol{X}(k)-\delta I)^{-1}\boldsymbol{e}(k)\\&+\frac{\gamma}{2}[\boldsymbol{X}(k)(\boldsymbol{X}^T(k)\boldsymbol{X}(k)-\delta I)^{-1}\boldsymbol{X}^T(k)\\&-I]\nabla\boldsymbol{\xi}_{l_0}(k)\end{aligned}\tag{11}$$

3. SIMULATION STUDY

In this section, the performance of the PZA-APA is compared with that provided by standard APA, ZA-APA, RZA-APA and l_0-APA. Four experiments have been carried out in a MATLAB environment to demonstrate their performance. The mean square deviation (MSD), which is the comparison metric used in simulation study, is defined as:

$$MSD(k)=\frac{1}{N+1}\sum_{i=1}^{N+1}[\mathbf{w}_i^0-\boldsymbol{w}^i(k)]^2\tag{12}$$

where $\mathbf{w}_i^0$ is the ith dimension coefficient of $\mathbf{w}_0$.

The length of the impulse response considered in the first three experiments is 16. The input signal employed in the modelling process is a white Gaussian noise with a variance of 1. A sparsity of 1/16 has been assumed for the first experiment, where the value of the seventh tap takes as 1 and all other coefficients set equal to 0. In the 2nd experiment, all the even taps are set to 1, while all the odd taps set equal to 0. For the 3rd experiment, a non-sparse system has been taken with all taps having a value of 1.

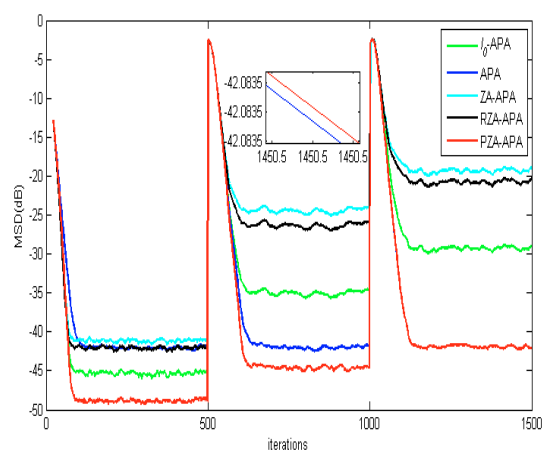

Fig. 2. *Convergence characteristics of various algorithms*

Fig. 2 shows the variation of MSD (average of 200 trials) with respect to iterations for this experiment. It is clear from the figure that the proposed PZA-APA algorithm outperforms the other algorithms considered in this Letter for the first 1000 iterations when we have used a sparse system. In the final 500 iterations where we have employed a non-sparse system, the performance of the PZA-APA algorithm is similar to that of the standard APA algorithm. The average steady-state MSDs computed over the last ten

iterations for each part are −41.38, −40.18, −42.42, −45.33 and −49.24 dB for APA, ZA-APA, RZA-APA, l_0-APA and PZA-APA algorithms, respectively, for the first 500 iterations. In general, all the four sparse APA algorithms studied are shown to provide improved modelling when the system is sparse in nature and among them PZA-APA offers the best modelling accuracy. The various simulations parameters used in this experiment are: μ = 0.25, γ = 0.0074 , N+1 = 16, ε = 200, $\delta = 1 \times 10^{-12}$, β = 4 and α = 5.

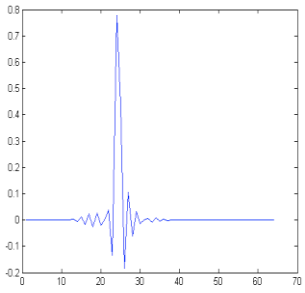

Fig. 3. *The impulse response of the feedback path*

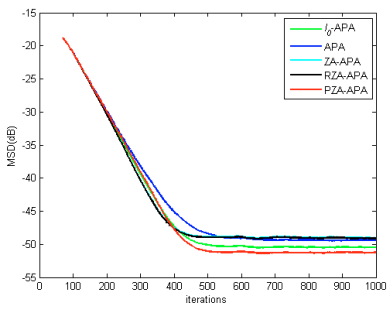

Fig. 3 *Experiment 2: Comparison of convergence characteristics*

In the fourth experiment, the length of the impulse response is 64. Fig. 3 shows the impulse response of the fourth experiment. It may be observed that a significant number of the coefficients in this impulse response are zeros, making it a sparse system. The input signal considered is same as that used in the previous experiment. The average MSDs computed as an average of last 100 iterations are −49.01, −49.85, −49.94, −50.66 and −51.95 dB for APA, ZA-APA, RZA-APA 1_0-APA and PZA-APA algorithms, respectively. The proposed PZA-LMS has provided improved modeling accuracy in comparison with other algorithms considered. The simulation parameters used are: μ = 0.25, γ = 0.0074 , N+1 = 64, ε = 200, $\delta = 1 \times 10^{-12}$, β = 4 and α = 20

4. CONCLUSION

A polynomial zero attracting affine projection algorithm is prosed in this work for sparse system identification. This algorithm uses a polynomial to approximate the l_0-norm in order to further improve the zero attraction capability of sparse adaptive algorithms for sparse system identification. Simulations have shown that the proposed algorithm has better convergence performance compared to other algorithms in sparse system identification.

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